Paper Note of Manifold Learning for Molecular ML

* (ii) Learning on topological surface and geometric structure for 3D molecular generation (Nature Computational Science 2023.10)

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1. Abstract

- Practical structure-specific three-dimensional molecule generations have started to emerge in recent years, **but most approaches treat the target structure as a conditional input to bias the molecule generation and do not fully learn the detailed atomic interactions that govern the molecular conformation and stability of the binding complexes.**
- Here, to address this challenge, we formulate a model, called SurfGen, that designs molecules **in a fashion closely resembling the figurative key-and-lock principle.**
- SurfGen comprises two equivariant neural networks, **Geodesic-GNN and Geoatom-GNN, which capture the topological interactions on the pocket surface and the spatial interaction between ligand atoms and surface nodes,** respectively.
- Its high sensitivity on the pocket structures enables an **effective generative-modelbased solution to the thorny issue of mutation-induced drug resistance.**

2. Introduction

- SBDD offers two main protocols: virtual screening，molecular generation。
- Therefore, **there is often no practical advantage for many existing generative models over virtual screening even if the generation process can theoretically access far more regions in the chemical space。**
- The study of interactions between protein pockets and small molecules is key to addressing these challenges.
- Recently, a new characterization approach has emerged in the field of protein–protein interaction (PPI) prediction, where proteins are modeled as abstract surfaces. **This approach takes into account the compatibility of protein–protein interface geometry and is able to capture intricate interface–interface interactions** [1] more directly due to the explicit modeling of protein topology.

[1] Gainza P, Sverrisson F, Monti F, et al. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning[J]. Nature Methods, 2020, 17(2): 184-192.

- Encouraged by the impressive performance of this surface representation for modeling PPI , **we adopt the protein surface channels as the protein representation for SBDD, embed relevant energy features on the surface and then perform the topological learning via our designed Geodesic-Graph Neural Network (GNN) framework.**
- We propose another framework called **Geoattn-GNN to learn this additional and crucial contribution of binding interactions.**
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    - SurfGen's two key components, Geodesic-GNN and Geoattn-GNN blocks, are constructed to learn topological and geometric interactions, respectively.
    - We conduct extensive experiments to prove that the molecules generated by SurfGen not only possess higher binding affinity to the protein pockets compared with the original ligands and **those generated by other state-of-the-art (SOTA) approaches but also form optimal geometric and energetic complementarities with the protein pockets.**
    - In addition, the results on real-world drug discovery cases demonstrate that SurfGen-generated molecules show the highest similarity to the experimental active molecules, and **exemplary molecule generation has been done to expand a focused compound library targeting coronavirus disease (COVID-19).**
    - Moreover, the experiments on **mutated targets demonstrate that SurfGen is sensitive enough to perceive the implications of critical mutations of a protein target, and noticeable mutation-induced impacts on the generated molecules can be observed.**

3. Results

- **The binding affinity of generated molecules should be used as a fundamental metric** for model evaluation.
- We have collected **some important therapeutic targets and the associated experimentally validated molecules to evaluate whether these models including SurfGen** could generate competitive drug candidates in real-world drug discovery projects.

3.1. Molecular generation on Test Set

- Dataset: CrossDock2020
- Autodock Vina: assess the affinity between small molecules and protein pockets.

3.2. Drug design on COVID-19 and other therapeutic targets

Table 2 | The binding energies and drug-likeness properties of the top-5 molecules designed for three real-world targets

	Actives	Decoys	GraphBP	Pocket2Mol	Morld	LigBuilder	SurfGen
			3CL top 5				
Docking energy (↓)	−7.599	−7.119	−7.479	−6.859	−6.180	−7.419	−8.540
QED	0.397	0.467	0.486	0.380	0.084	0.590	0.418
SA	0.696	0.703	0.7374	0.742	0.306	0.704	0.752
Lipinski	4.000	4.667	5.000	5.000	3.200	4.400	5.000
Log*P*	1.095	0.061	0.0976	−0.799	6.786	−0.297	−0.332
			AKT1 top 5				
Docking energy (↓)	−11.940	−10.340	−9.059	−9.600	−8.020	−9.720	−12.62
QED	0.264	0.506	0.601	0.745	0.027	0.647	0.456
SA	0.266	0.314	0.454	0.264	0.300	0.712	0.422
Lipinski	4.200	4.800	5.000	5.000	2.400	5.000	5.000
Log*P*	2.475	1.208	1.417	0.967	9.252	1.720	1.748
			CDK2 top 5				
Docking energy (↓)	−11.020	−9.920	−9.720	−9.300	−7.439	−9.660	−12.21
QED	0.407	0.563	0.581	0.667	0.178	0.646	0.748
SA	0.362	0.316	0.586	0.336	0.461	0.666	0.776
Lipinski	4.600	5.000	4.600	5.000	4.200	5.000	5.000
Log*P*	−0.893	0.821	1.765	0.397	4.510	1.658	0.756

All technical terms are discussed in the footnote of Table 1.

3.3. Mutants conditional molecular generation

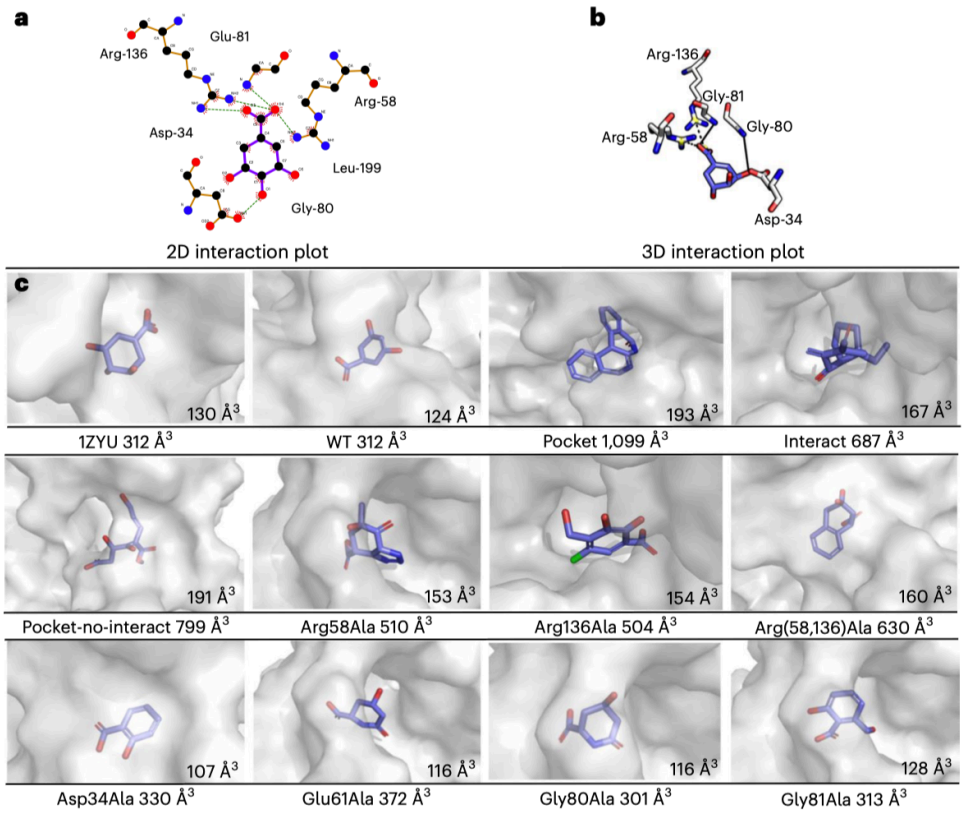
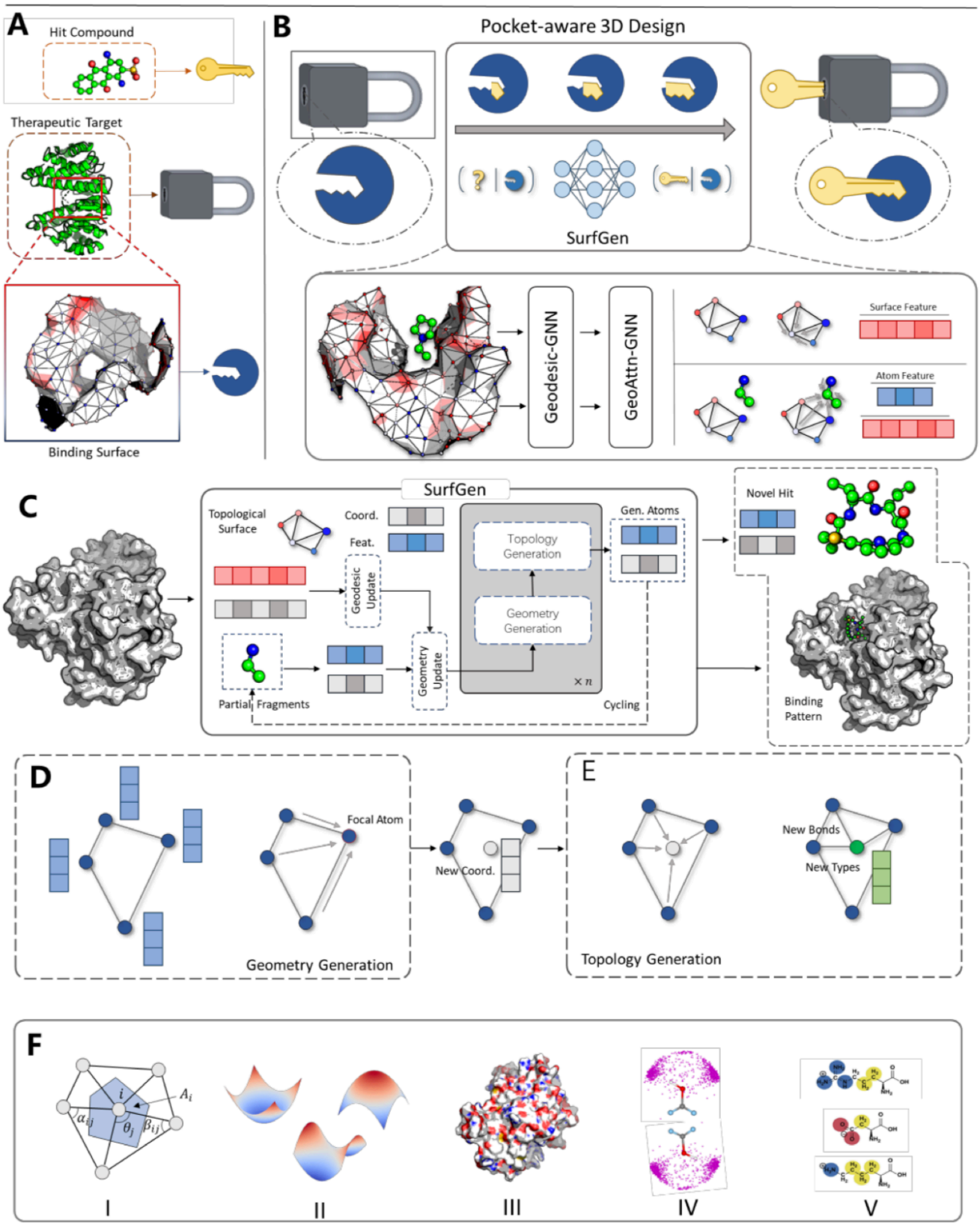


Fig. 3 | Conditional generated molecules on shikimate kinase mutants. **a**, Two-dimensional interaction pattern for the wild-type crystal structure. The important residues are highlighted by the residue name ID. Carbon, black; oxygen, red; nitrogen, blue. **b**, Three-dimensional interaction pattern on the wild-type crystal structure. The important residues are also shown. Carbon, purple; chlorine, green; oxygen, red; nitrogen, blue. **c**, Visualization examples of the generated samples toward the mutants. The volume at the bottom of image represents the volume of the molecules and the mutation notations are followed by the volume of the pocket.

4. Discussion

5. Methods



- Design **Geodesic-GNN, a topological learning framework combined with equivariant constraint, to learn on curved surfaces with topological attention.**

- Three geometric and three energetic features are extracted and attached to the nodes of the pocket surface, where the geometric features include **Gaussian curvature, mean curvature and shape index**; the energetic features include **Poisson–Boltzmann charge , hydrogenbond donors and hydropacy.**
- **Gaussian curvature, mean curvature and shape index, Poisson-Boltzmann charge, hydrogen bond donors and hydrophilicity.**

- Design Geoattn-GNN based on the geometric attention mechanism for **learning the matching laws between atoms in molecule and pockets.**
- In SurfGen, **the autoregressive protocol is adopted so that the generated atoms can fully interact with pocket environment and thus learn geometry matching more efficiently.**
  - **Geometry generation**
    To satisfy the equivariance requirement of the geometry generation, that is, predict the coordinate of the next atom, the techniques from VP and GVP were adopted here to construct this module. In addition, the negative log likelihood (NLL) loss function was used to fit a 3D multivariate Gaussian distribution:

$$p(\Delta \mathbf{r}_i) = \sum_{k=1}^{K} \pi_i^{(k)} \mathcal{N}(\mu_i^{(k)}, \Sigma_i^{(k)}) \tag{1}$$

$$p(\Delta \mathbf{r}_i + \mathbf{r}_j) = \sum_{k=1}^{K} \pi_i^{(k)} \mathcal{N}(\mu_i^{(k)} + \mathbf{r}_j, \Sigma_i^{(k)}) \tag{2}$$

$$(s_\pi, V) = GVP_p(s_\mathbf{h}, V_\mathbf{h}) \tag{3}$$

$$(s_\mu, V_\mu) = GVP_\mu(s_\mathbf{h}, V_\mathbf{h}) \tag{4}$$

$$(s_\mu, V_\mu) = GVP_\epsilon(s_\mathbf{h}, V_\mathbf{h}) \tag{5}$$

$$\mu = \mu_v \tag{6}$$

$$\Sigma = MLP_\Sigma(s_\Sigma) \tag{7}$$

$$\pi = MLP_{\pi_2}(MLP_{\pi_1}(s_\pi) + ||V_\pi||_2) \tag{8}$$

where $\mathbf{r}_i$ is the Cartesian coordinate of i_{th} generated atom; $\mathcal{N}$ is the Gaussian distribution, π, μ, Σ, and K are the combination coefficient, mean, variance, and the number of components of the Gaussian mixed density network; $(s_\mathbf{h}, V_\mathbf{h})$ is the scalar and vector feature of hidden interaction space.

- Topology generation
  Knowing the spatial relationship between the next atom and other existing atoms, the atomic type and bonding relationship of the new atom can be determined. However, considering that the distribution of atom types inside the pocket is influenced by the chemical environment, SurfGen utilizes the structural k nearest neighbor (kNN) aggregation mechanism in the topology-generation module.

$$(s_{m(i)}, V_{m(i)}) = \sum_{k \in kNN(i)} \text{Message}(s_{h(k)}, V_{h(k)}, e_{ki}, U_{ki}) \tag{9}$$

$$p_{a_i} = MLP_{a_t 2}(MLP_{a_2}(s_{m(i)}) + ||V_{m(i)}||_2) \tag{10}$$

$$e'_{ij}, U'_{ij} = GVP_{bp1}([s_i||s_{ij}||e_{ij}], [||V_{ij}||V_{ij}, U_{ij}]) \tag{11}$$

$$e''_{ij}, U''_{ij} = \text{Attention}(e'_{ij}, U'_{ij}) + (e'_{ij}, U'_{ij}) \tag{12}$$

$$e_{ij}^{type} = MLP_{bp_2}(MLP_{bp_1}(e'_{ij}) + ||U''_{ij}||_2) \tag{13}$$

where Message is the message-passing framework; $(s_{h(k)}, V_{h(k)}, e_{ki}, U_{ki})$ are scalar and vector features of atoms and edges; p_{a_i} is the probability of types of next atom; Attention is triangle attention module; and e_{ij}^{type} is the bonding relationship between the next atom and other atoms.